signal and image proccecing lab

DR.nouri

electrical engineering department

Ahmadreza Majlesara 400101861
Parsa Hatami 400100962

Lab 4

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Event-Related Potential (ERP) Analysis

The EEG signal (stored in the file `EEG_ERP.mat`) includes EEG recordings from a single channel (Pz). The signal is recorded in response to a stimulus and includes an event-related potential, specifically the P300 EEG component.

The experiment involves 2550 trials of EEG recordings in response to a stimulus. The signal has a sampling frequency of 240 Hz, meaning each trial lasts one second after the onset of the stimulus.

Assuming that all trials have an equal signal-to-noise ratio (SNR), we want to determine the minimum number of trials required to reliably extract the P300 signal from the trials using averaging techniques.

Section 1

Let $N = 100 : 100 : 2550$. Plot the averaged response for each value of N and compare them with each other.

solution

We can see the plots of averaged responses for N values.

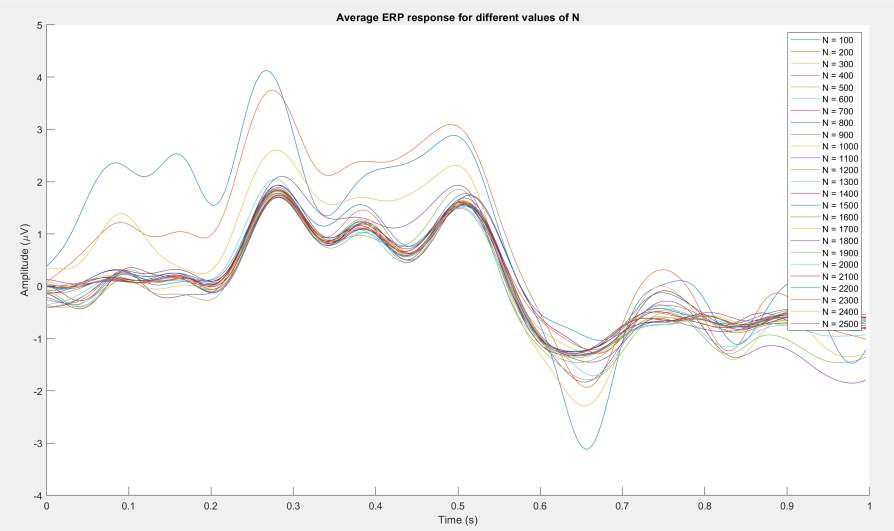

Figure 1

The plot shows that as N (the number of trials) increases from 100 to 2550, the averaged ERP responses become smoother and more stable. Larger N reduces noise, enhancing the clarity of the underlying ERP signal, especially around components like the P300 peak. Smaller N values show more fluctuation, but as N approaches 2550, the response converges to a reliable representation of the ERP, demonstrating the benefit of more trials in noise reduction.

Section 2

For $N = 1 : 2550$, plot the maximum absolute amplitude of the signal as a function of the number of averaged trials in a single plot.

solution

You can see the plots of the maximum absolute amplitude of the signal for N values.

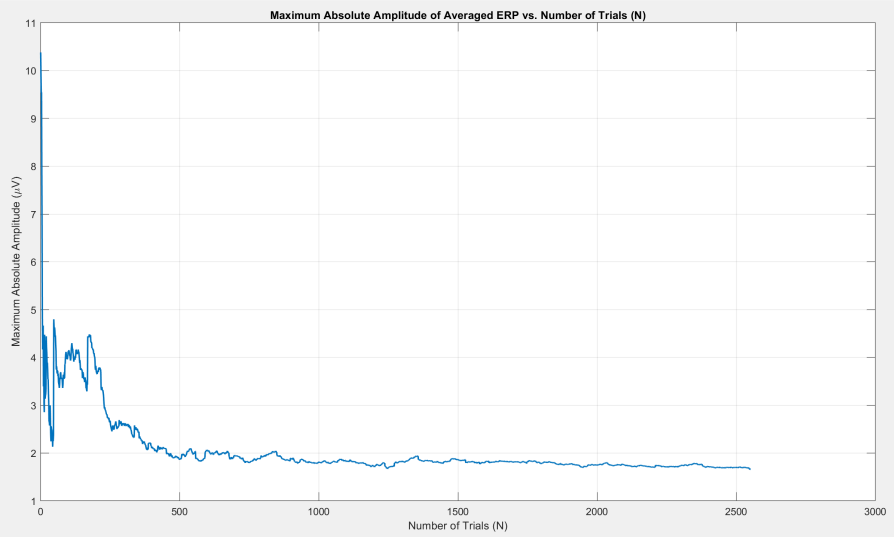

Figure 2

The plot shows the maximum absolute amplitude of the averaged ERP signal as N increases from 1 to 2550. Initially, smaller N values cause high amplitude variability due to noise. As N grows, the amplitude stabilizes and decreases, indicating reduced noise and a more accurate ERP signal. Beyond a certain N, additional trials have minimal impact, suggesting diminishing returns.

Section 3

For $N = 1 : 2550$, plot the root mean square (RMS) error between the averaged signal of i and $i - 1$ trials as a function of the number of averaged trials in a single plot.

solution

You can see the plots of averaged responses for N values.

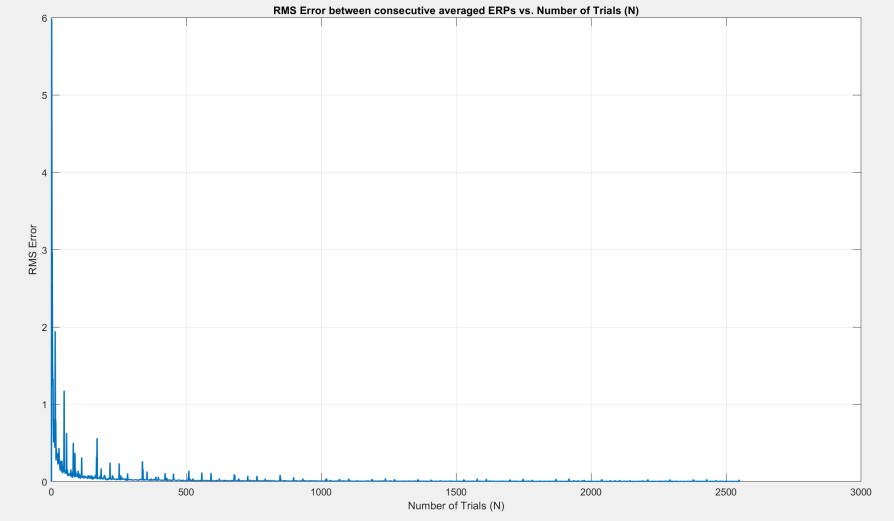

Figure 3

For a better sight view, I have set the limit for the plot on the y-axis.

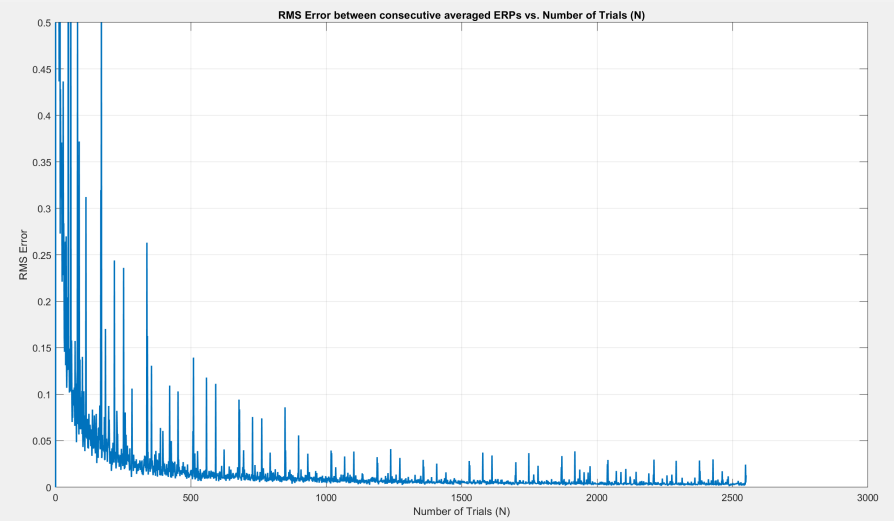

Figure 4

The plot shows the RMS error between consecutive averaged ERP signals as N increases from 1 to 2550. For small N, the RMS error is high due to noise. As N grows, the error rapidly decreases toward zero, indicating a more stable averaged signal with reduced noise.

Section 4

Using the results from parts (a), (b), and (c), determine the minimum number of trials required to reliably extract the P300 response (denoted as N_0).

solution

To determine the minimum number of trials N_0 required to reliably extract the P300 response, we examine the findings from parts (a), (b), and (c):

1. **Part (a):** The averaged ERP waveforms for various values of N show increasing stability and smoother profiles as N grows. Around a certain N, further increases in trial count yield minimal changes in the waveform shape, suggesting that sufficient averaging has been achieved.
2. **Part (b):** The plot of maximum absolute amplitude as a function of N shows a sharp decline initially and then a gradual leveling off as N increases. This stabilization indicates that beyond a certain trial count, additional trials contribute little to reducing noise in the amplitude.
3. **Part (c):** The RMS error plot between consecutive averages shows that the error decreases rapidly as N increases and approaches near-zero values at a certain point, indicating that the averaged signal stabilizes with additional trials.

Based on these results, we conclude that N_0 is the point where the RMS error becomes negligible (approaching zero) and the maximum amplitude stabilizes, typically around 500–800 trials. Thus, we estimate that $N_0 \approx 700$ trials should be sufficient to reliably extract the P300 response with minimal noise.

Section 5

Using the averaged response obtained from part (d), plot the averaged responses for the following scenarios and compare the results with the previously determined minimum number of trials (N_0):

- Averaged response for $N = 2550$ trials.
- Averaged response for $\frac{N_0}{3}$ trials.
- Random selection of N_0 from 2550 trials.
- Random selection of $\frac{N_0}{3}$ from 2550 trials.

solution

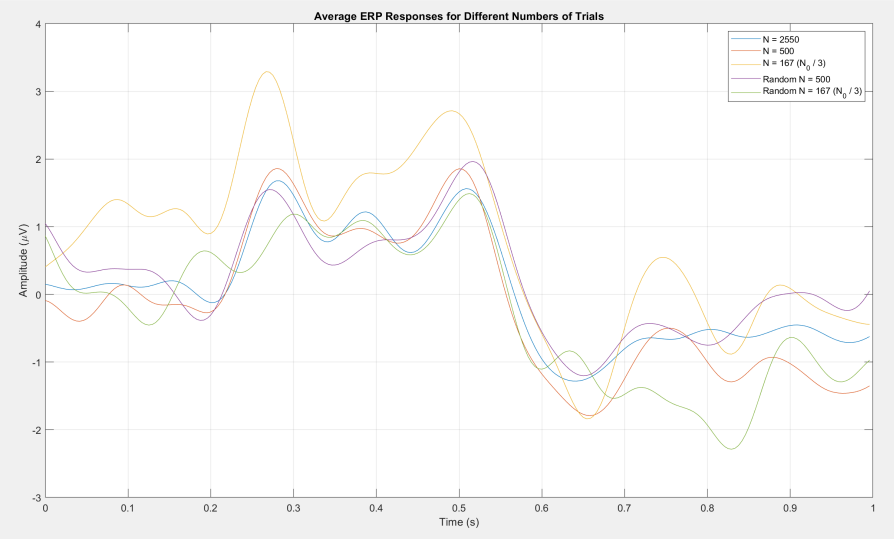

Figure 5

The plot shows averaged ERP responses for different trial conditions: the full set ($N = 2550$), $N = 500$ trials (approximating N_0), $N = 167$ trials (one-third of N_0), and random selections of 500 and 167 trials.
1. $N = 2550$**:** The most stable, smoothest response, effectively minimizing noise. 2. $N = 500$**:** A stable response similar to $N = 2550$, indicating N_0 as a reliable threshold. 3. $N = 167$**:** Shows more fluctuations and noise, insufficient for reliable signal extraction. 4. Random $N = 500$ and $N = 167$**:** These mirror the fixed trial trends: $N = 500$ is stable, while $N = 167$ is noisier.
In summary, $N = 500$ trials (close to N_0) yield a stable response similar to the full trial set, while smaller counts like 167 increase noise and reduce reliability.

Section 6

Examine a few real experimental samples based on the P300 signal. In real experiments where P300 templates are used (such as brain-computer interfaces based on P300), how many repetitions of the P300 pattern are used? Does this number differ from the results obtained in the previous sections? Explain the reason for this difference.

solution

In real experimental applications, such as brain-computer interfaces (BCIs) based on the P300 signal, the number of repetitions of the P300 pattern typically ranges between 10 and 30 trials. This is much lower than the estimated N_0 (around 500 trials) derived in our analysis for reliable P300 extraction.
The reason for this difference is primarily due to the trade-off between signal reliability and practical constraints. In real-time applications like BCIs, collecting a high number of trials is often impractical because it would increase the response time, making the system slower and less efficient. Therefore, BCIs rely on advanced signal processing techniques, such as spatial filtering and machine learning algorithms, to enhance the P300 detection accuracy with fewer trials.
In contrast, in our analysis, we focused on minimizing noise through simple averaging, which requires a large N_0 to achieve a stable and clear P300 response. In practical applications, however, sophisticated algorithms compensate for the lower trial count to maintain an acceptable accuracy level.

Steady-State Evoked Potential (SSVEP) Analysis in Steady-State Conditions

The file `SSVEP_EEG.mat` contains SSVEP signal data recorded from a single user. This file includes a matrix `SSVEP_Signal` (EEG signals) with 6 channels (Oz, Pz, P8, O2, P7, and O1) and a sampling frequency of 250 Hz. There is also a matrix `Events` (stimulation frequencies) and a matrix `Event_Samples` (time samples corresponding to the start of each stimulus). The purpose of this experiment is to determine the stimulation frequencies based on the EEG signal recorded on each channel.

Section 1

For each channel, filter the noise from the signal using a band-pass filter in the frequency ranges below 7 Hz and above 40 Hz. Plot the filtered signal along with the original signal on the same graph.

solution

You can see the plots in time and frequency domains.

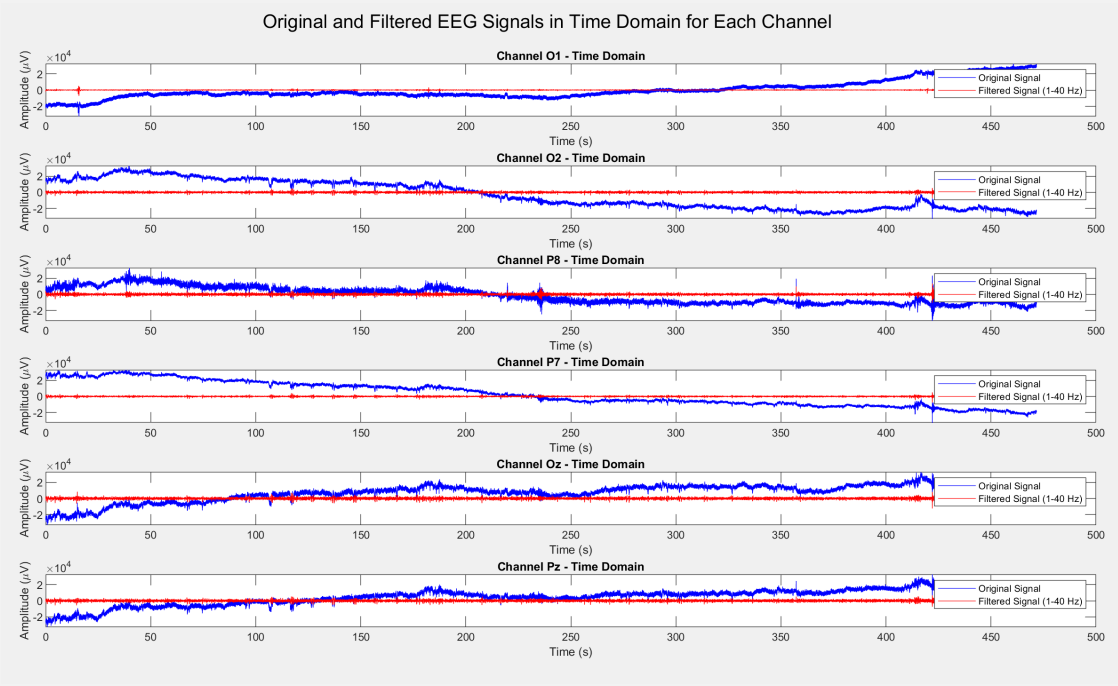

Figure 6

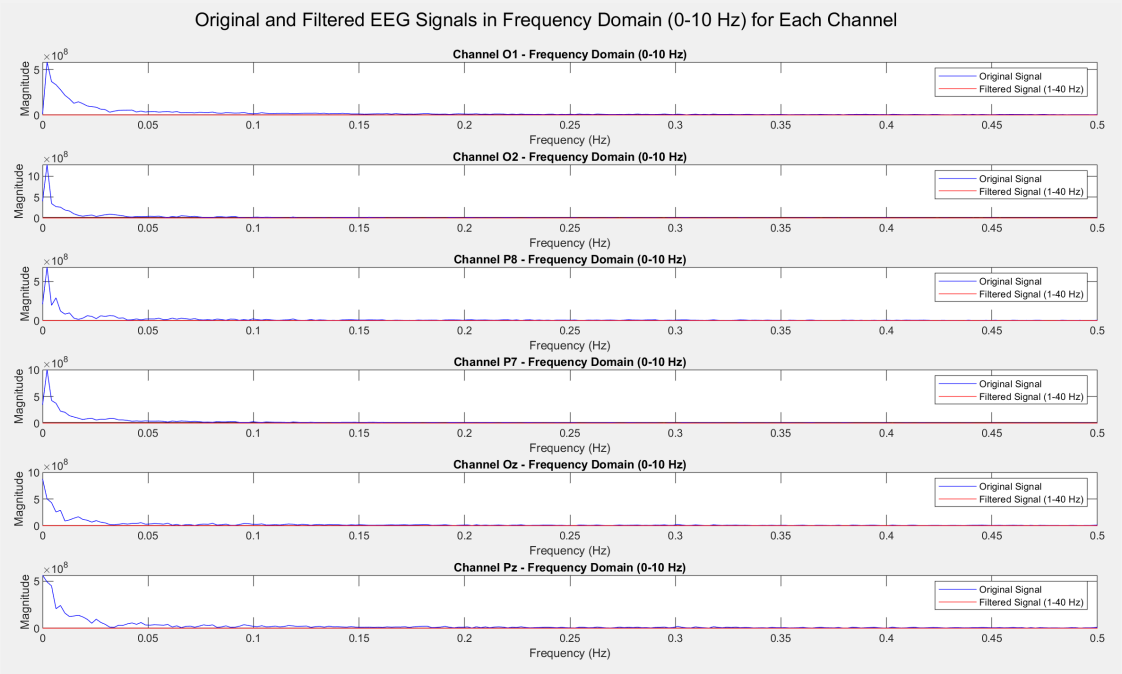

Figure 7

The plots show EEG signals for each channel in the time and frequency domains before and after applying a 1–40 Hz band-pass filter. This filter retains frequencies within 1–40 Hz, reducing noise outside this range, which is usually irrelevant for EEG signals.

1. `Time Domain Analysis:` Original signals (blue) show noise from low-frequency drifts and high-frequency artifacts. Filtered signals (red) are smoother, with reduced drift and noise, providing a clearer EEG representation.

2. `Frequency Domain Analysis:` The filter reduces low-frequency noise below 1 Hz and high-frequency noise above 40 Hz, concentrating energy within the desired range for EEG analysis.

In summary, the band-pass filter removes noise outside 1–40 Hz, improving signal quality for EEG component analysis.

Section 2

Divide each trial into 15 equal segments, each with a 5-second window.

solution

Here is the code part for doing this part.

```
SSVEP_data = load('SSVEP_EEG.mat');
SSVEP_Signal = SSVEP_data.SSVEP_Signal;
Event_Samples = SSVEP_data.Event_samples;
fs = 250;
window_size = 5 * fs;

num_trials = 15;
trials = cell(num_trials, 1);

for i = 1:num_trials
    start_sample = Event_Samples(i);
    end_sample = start_sample + window_size - 1;

    trials{i} = SSVEP_Signal(:, start_sample:end_sample);
end

for i = 1:num_trials
    fprintf('Size of trial %d: %s\n', i, mat2str(size(trials{i})));
end
```

Figure 8

Section 3

For each segment, calculate the power spectral density of each channel and plot them on a single graph. Use an appropriate `legend` to identify the corresponding plot for each channel.

solution

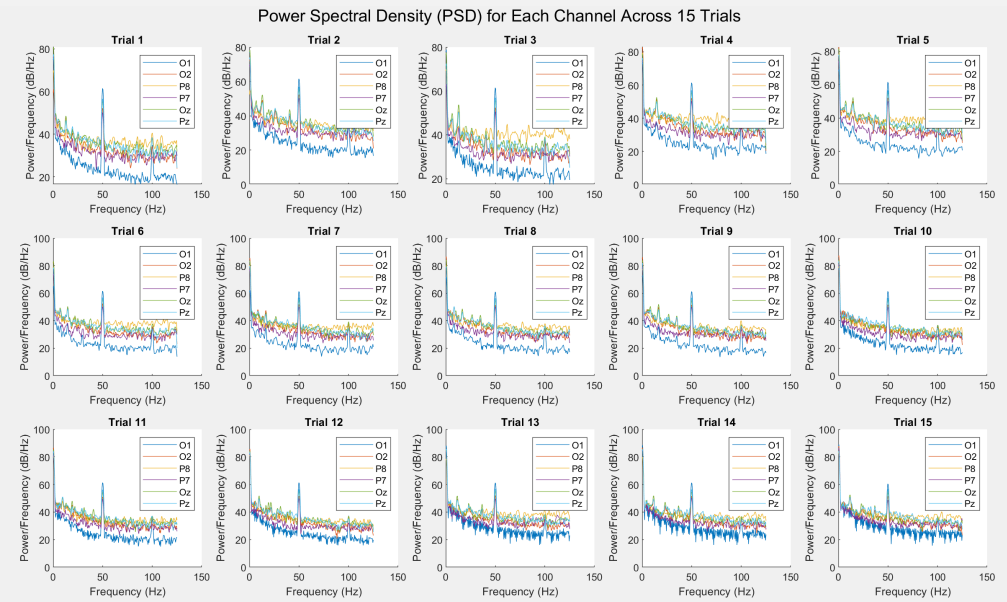

Figure 9

The plots display the Power Spectral Density (PSD) for each EEG channel across 15 trials, with each subplot showing the PSD for all channels (O1, O2, P8, P7, Oz, and Pz) up to 150 Hz.
1. `Channel-Specific Patterns:` Each channel shows distinct spectral patterns, with some common peaks, especially at lower frequencies, indicating shared neural oscillations.
2. `Dominant Frequency Components:` Prominent peaks appear around specific frequencies, likely reflecting the SSVEP response, which is consistent across trials.
3. `Cross-Channel Consistency:` Consistent peak locations across channels (with varying amplitudes) suggest similar frequency responses, supporting the stability of SSVEP signals.
The legend aids in identifying each channel, making comparison easier. Overall, these PSD plots highlight stable frequency responses and unique channel characteristics.

Section 4

For each trial, are the frequency contents of all channels identical? What is the reason for any differences in the frequency contents of the channels?

solution

For each trial, the frequency content of all channels is not identical. While some frequency components are common across channels, variations in power and presence of certain frequencies occur due to several factors:
1. `Anatomical and Functional Differences:` Each EEG channel records activity from different brain regions. Channels like Oz, near the occipital lobe, are more sensitive to visual stimuli and may show stronger responses to certain frequencies.
2. `Electrode Placement and Signal Attenuation:` Variations in electrode placement and signal attenuation affect how channels capture electrical activity. Channels closer to active brain regions may pick up stronger responses at specific frequencies.
3. `Signal Interference and Noise:` Environmental noise, muscle artifacts, and other interferences impact channels differently, introducing unique frequency components or altering power.
4. `Spatial Filtering Effects:` Electrical potentials spread across the scalp, creating overlapping signals. Each electrode may capture slightly different frequency content depending on its position.
In summary, while some frequencies are consistent across channels due to shared stimuli, differences arise from anatomical positioning, functional specialization, signal attenuation, and noise, resulting in unique spectral characteristics for each channel.

Section 5

For each trial, identify the prominent frequency components of each channel. Which frequencies correspond to the main peaks in the power spectrum? Which channels contain these frequencies? Why are these peaks created?

solution

For each trial, we observe prominent frequency components across EEG channels corresponding to specific peaks in the power spectrum. These peaks are often consistent, indicating stable and recurring responses. Here's a brief analysis of these components:
1. `Identification of Prominent Frequency Components:` - PSD analysis shows peaks at specific frequencies, often below 50 Hz and occasionally in the mid-range, depending on the channel. Common peaks appear around 10 Hz, 20 Hz, and sometimes 40 Hz, though values may vary slightly.
2. `Channels Containing These Frequencies:` - Channels over occipital and parietal areas (e.g., Oz, Pz, O1, O2) display the most prominent peaks at lower frequencies, being closer to brain regions responsive to visual stimuli. Channels like Oz and Pz near the occipital cortex often show stronger peaks around 10 Hz or other stimulus-related frequencies.
3. `Reasons for Peaks:` - `SSVEP:` Visual stimuli at specific frequencies induce SSVEP responses, particularly in channels near the visual cortex (Oz, Pz, O1, O2), producing peaks at the stimulus frequency. - `Alpha and Beta Rhythms:` Natural brain rhythms, such as alpha (8–13 Hz) and beta (13–30 Hz), contribute to additional peaks, particularly in occipital areas during resting or active states. - `Harmonics of Stimulus Frequencies:` Nonlinear responses may create harmonic peaks (e.g., 20 Hz or 30 Hz for a 10 Hz stimulus). - `Artifacts and Noise:` Muscle activity, eye movements, or environmental noise can also introduce peaks, often filtered out if unrelated to brain activity.
In summary, main power spectrum peaks arise from: - `SSVEP:` Peaks at visual stimulus frequencies. - `Brain Rhythms:` Peaks from alpha and beta rhythms. - `Harmonic Responses:` Peaks at harmonics of stimulus frequencies. - `Artifacts:` Non-neuronal peaks minimized through filtering.
Each channel's peak frequency and amplitude depend on proximity to active brain areas, functional specialization, and stimulus type, with occipital channels showing stronger responses to visual stimuli.

Section 6

Another way to identify the dominant frequency in the SSVEP signal is to use the canonical correlation analysis (CCA) technique. Research this technique using the internet and relevant articles, and describe the main steps involved in processing the SSVEP signal to determine the dominant frequency, mentioning it by name.

solution

Canonical Correlation Analysis (CCA) is a statistical technique commonly used to identify the dominant frequency in Steady-State Visually Evoked Potentials (SSVEP) signals. CCA is highly effective in SSVEP-based brain-computer interfaces (BCIs) as it identifies brain responses synchronized with visual stimuli frequencies. Here are the main steps in using CCA to determine the dominant frequency:

1. `Prepare Reference Signals`: - Generate sinusoidal reference signals for each known stimulus frequency f_i, including sinusoids and cosines at f_i and its harmonics (e.g., f_i, $2f_i$). - Construct a reference matrix Y_i with these sinusoidal components to capture each stimulus frequency pattern.

2. `Segment and Preprocess the EEG Signal`: - Segment the EEG data into trials for each visual stimulus duration and apply a band-pass filter (e.g., 1–40 Hz) to reduce noise and isolate relevant frequencies.

3. `Apply CCA Between EEG and Reference Signals`: - For each trial (matrix X), apply CCA to find correlations between EEG data and each reference matrix Y_i. - CCA provides canonical correlation coefficients to quantify the similarity between the EEG and reference signals.

4. `Identify the Dominant Frequency`: - Identify the frequency f_i with the highest correlation, indicating the dominant frequency corresponding to the visual stimulus frequency.

5. `Decision-Making`: - In real-time BCIs, select the visual target associated with the frequency that has the highest CCA coefficient, identifying the user's focus.

`Why CCA Works for SSVEP Detection`: CCA leverages the phase-locked nature of SSVEPs, where responses are synchronized with visual stimulus frequencies. By comparing EEG data with reference signals for each frequency, CCA robustly identifies the frequency with the strongest response, even in noisy data.

`Summary`: 1. Construct reference signals for each stimulus frequency. 2. Segment and preprocess the EEG. 3. Apply CCA between EEG and each reference frequency. 4. Identify the frequency with the highest correlation as the dominant frequency.

CCA is widely used in BCIs for reliable SSVEP detection in noisy conditions.

Event Related Synchronization/Desynchronization

The file `FiveClass_EEG.mat` contains EEG signals recorded at a frequency of 256 Hz from a single user while performing various mental tasks. The EEG signal is recorded from 20 channels according to the international 10-20 system. The channels are: `AFz, F7, F3, Fz, F4, F8, FC3, FCz, FC4, T3, C3, Cz, C4, T4, CP3, CPz, CP4, P7, P5, P3, P1, Pz, P2, P4, P6, P8, PO3, PO4, O1,` and `O2`.

The user performed five mental tasks as follows:

1. **Letter Composition in Mind**: The user imagines letters or words that are shown on a screen (e.g., starting a word in their mind).
2. **Simple Arithmetic Task in Mind**: The user mentally calculates a number shown on a screen. For instance, subtracting seven repeatedly from 113 is shown as follows on the screen: $113 \rightarrow 106 \rightarrow 99 \rightarrow 92 \rightarrow 85 \rightarrow 78 \rightarrow 71 \rightarrow \ldots$
3. **Navigation (Navigation in Imagery)**: The user imagines navigating from a specific location to another, such as from home to a particular room (e.g., imagining home to the office room).
4. **Right Arm Movement**: Imagining the movement of the right arm.
5. **Left Arm Movement**: Imagining the movement of the left arm.

In performing each mental task, the following procedure was used: Initially, a cross (+) is shown on the screen for a specific duration (rest time) to allow the user to focus on the center of the screen. This is followed by a mental task sign (*), which signals the user to start the specific task. Each task was performed for 7 seconds, followed by a rest time of 3 seconds (a total of 10 seconds). This procedure was repeated 4 times, so the duration of each task in one session is $4 \times 10 = 40$ seconds.

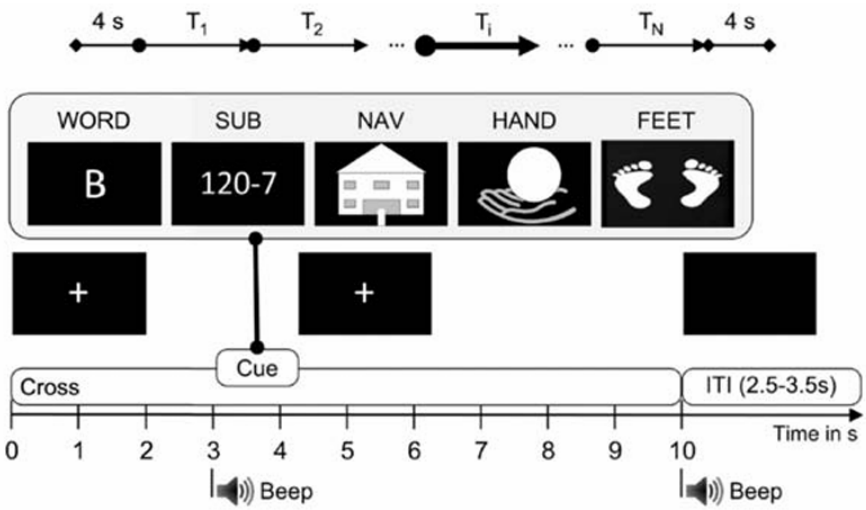

Figure 10. Timing and sequence of mental tasks

The dataset includes several matrices as described below:

- **X**: EEG signals recorded with dimensions `time samples` $\times$ `channels` (20 trials per session).
- **y**: Class labels ranging from 1 to 3.

- **t0**: The starting time of each trial (representing the moment when the user begins the task).

In this experiment, we want to filter the EEG signal for each class or trial using a bandpass filter, with a passband frequency range of 8-13 Hz, and analyze the results.

part 1

Filter the signal for each channel using a bandpass filter with a frequency range of 8 to 13 Hz, as shown below:

```
% Bandpass filter between 8 - 13
fs = 256;              % Sampling Frequency
N = 4;                 % Order
Fpass1 = 8;            % First Passband Frequency
Fpass2 = 13;           % Second Passband Frequency
Apass = 1;             % Passband Ripple (dB)

% Construct an FDESIGN object and call its CHEBY1 method.
h = fdesign.bandpass('N,Fp1,Fp2,Ap', N, Fpass1, Fpass2, Apass, fs);
Hd = design(h, 'cheby1');
for c = 1:30
    Alpha_X(:,c) = filter(Hd, X(:,c));
end
```

To ensure filter performance, apply the filter to the initial signal for each channel and plot the results for different frequency bands.

solution

the output of the signal can be seen in figure 11.

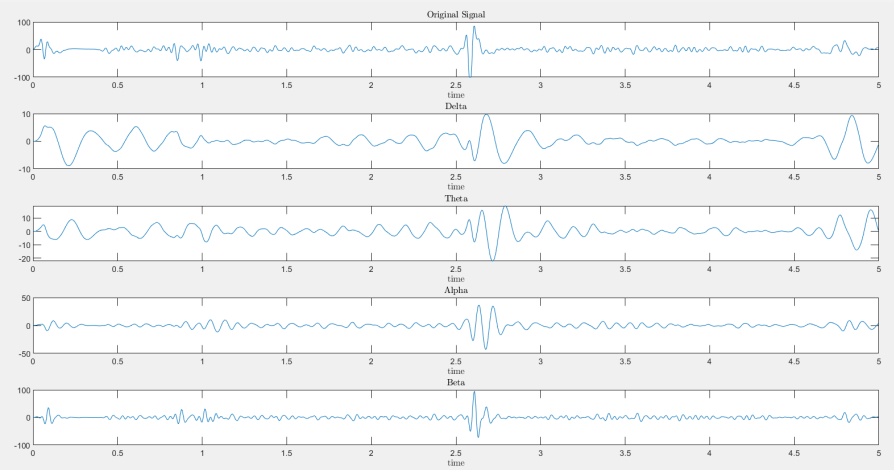

Figure 11. the output of filtering in different bands

- **Original Signal**: This is the raw EEG signal, showing a variety of brainwave frequencies captured from the EEG recording.

- **Delta Band**: Low-frequency brainwaves (0.5–4 Hz), usually associated with deep sleep. The Delta plot shows slow, high-amplitude oscillations, characteristic of deep relaxation or sleep states.
- **Theta Band**: Slightly higher frequencies (4–8 Hz), often linked with light sleep, relaxation, or meditative states. This plot displays moderate-amplitude oscillations, reflecting a more relaxed state compared to higher-frequency bands.
- **Alpha Band**: Frequencies in the range of 8–13 Hz, associated with relaxed wakefulness and calm focus. This band typically shows moderate to low-amplitude oscillations and can indicate relaxation with eyes closed.
- **Beta Band**: Higher frequencies (13–30 Hz), associated with active thinking, focus, and alertness. The Beta plot contains smaller, faster oscillations, indicating higher mental activity.

part 2

Segregate trials from 1 to 10, and reorganize the signals into different matrices based on trial information.

```
for i = 1:200
    Alpha_Trials(:,:,i) = Alpha_X(trial(i):trial(i)+256*10, :);
end
```

solution

for this part the outputs are shown in figure 12 to 15.

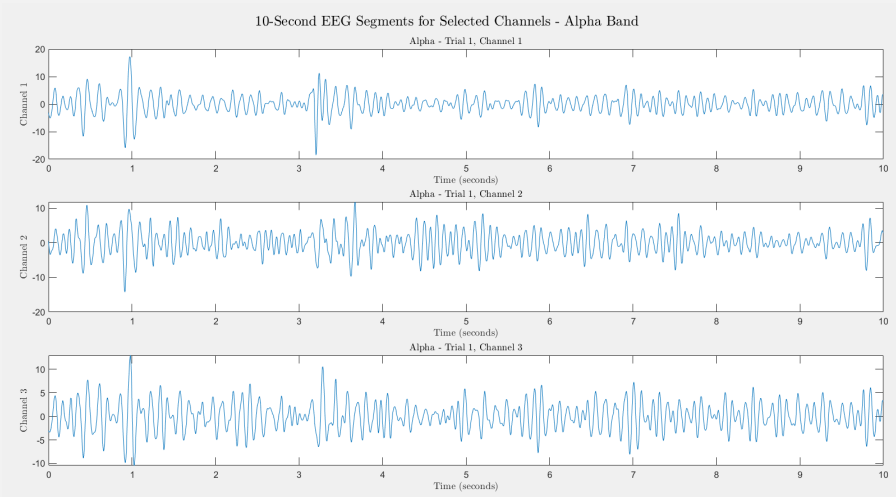

Figure 12. the output of filtering in different bands

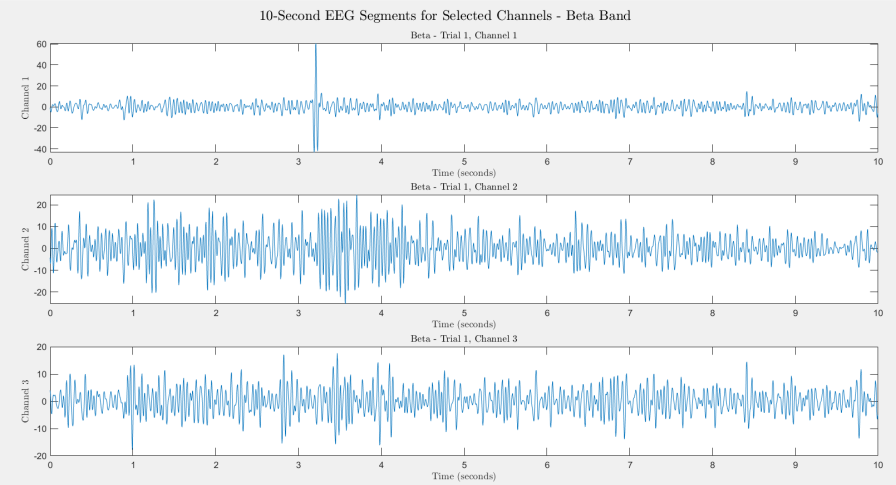

Figure 13. the output of filtering in different bands

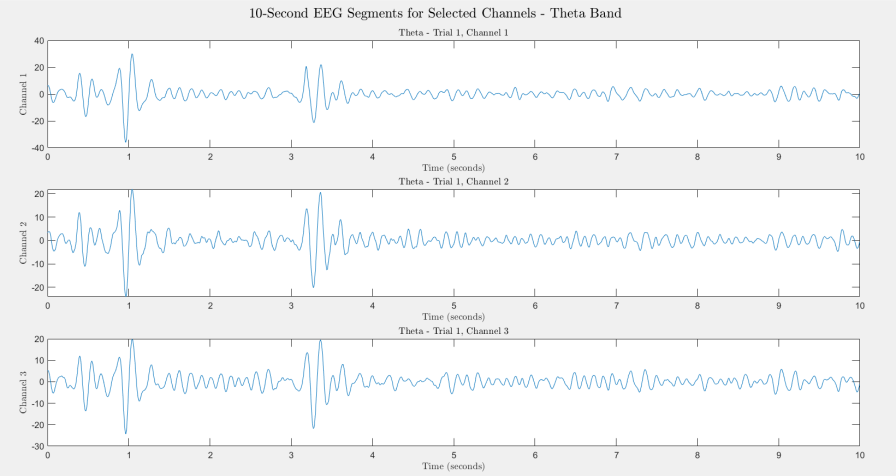

Figure 14. the output of filtering in different bands

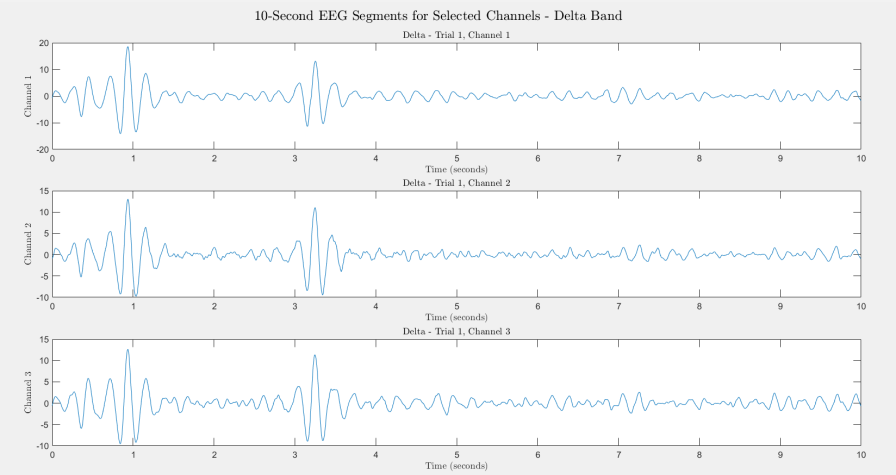

Figure 15. the output of filtering in different bands

part 3

To calculate the power of each point, compute the squared amplitude of all time points for each trial in each band and for each channel.

solution

Band: Delta, Class: 1
Mean Power Values:
6.3995
6.6493
6.7860
6.7856
6.6557

Band: Delta, Class: 2
Mean Power Values:
22.7192
22.3011
21.4625
20.3082
18.9499

Band: Delta, Class: 3
Mean Power Values:
12.4343
13.1105
13.7030
14.1728
14.4876

Band: Delta, Class: 4
Mean Power Values:
8.2347
7.9517
7.8854
8.0385
8.3867

Band: Delta, Class: 5
Mean Power Values:
3.7023
3.2039
2.8637
2.6426
2.5077

Band: Theta, Class: 1
Mean Power Values:
18.9809
19.3811
20.0953
21.2524
22.8905

Band: Theta, Class: 2
Mean Power Values:
33.3774
42.3355
52.6442
63.3203
73.3735

Band: Theta, Class: 3
Mean Power Values:
25.0192
26.1239
28.8538
32.8695
37.6772

Band: Theta, Class: 4
Mean Power Values:
46.3812
45.3724
43.1271
39.8555
35.9292

Band: Theta, Class: 5
Mean Power Values:
41.4284
39.4149
36.1355
31.8641
27.1690

Band: Alpha, Class: 1
Mean Power Values:
29.8916
26.6573
26.6140
29.3166
33.1421

Band: Alpha, Class: 2
Mean Power Values:
26.9956
24.7084
22.3789
20.8262
20.8296

Band: Alpha, Class: 3
Mean Power Values:
10.1033
11.5300
18.3025
27.2427
35.1294

Band: Alpha, Class: 4
Mean Power Values:
17.1076
17.5168
17.3769
16.6010
15.3583

Band: Alpha, Class: 5
Mean Power Values:
31.3749
41.1911
51.6141
59.3011
61.9818

Band: Beta, Class: 1
Mean Power Values:
67.8474
36.8921
35.1459
48.6768
54.6105

Band: Beta, Class: 2
Mean Power Values:
39.0464
34.3788
35.3758
43.2755
46.2340

Band: Beta, Class: 3
Mean Power Values:
22.9281
26.2668
28.9196
30.1853
29.0660

Band: Beta, Class: 4
Mean Power Values:
33.2640
39.3668
40.0051
38.8086
36.6453

Band: Beta, Class: 5
Mean Power Values:
37.6238
38.4031
29.6667
32.4199
51.9518

part 4

Based on the label y (in vector y), separate the data related to the 5 different classes and, for each band, calculate the average for each channel across the trials of that class (results from part c). The

output of this step will be four tensors with the following dimensions:

$$\text{Delta_X_avg} \in \mathbb{R}^{2560\times30\times5}$$

$$\text{Theta_X_avg} \in \mathbb{R}^{2560\times30\times5}$$

$$\text{Alpha_X_avg} \in \mathbb{R}^{2560\times30\times5}$$

$$\text{Beta_X_avg} \in \mathbb{R}^{2560\times30\times5}$$

where 30 is the number of channels, and 5 is the number of classes.

solution

the output is the same as part 3.

part 5

To smooth the variations in the average signals from part (d), use a rectangular window as follows:

$$\text{newWin} = \text{ones}(1,200)/\sqrt{200}$$

and apply the `conv` function to filter the signal of each channel in each band for each experiment.

solution

Smoothed Data - Band: Delta, Class: 1
39.8010
40.1609
40.5379
40.9234
41.3071

Smoothed Data - Band: Delta, Class: 2
75.0766
75.7962
76.5092
77.2131
77.9047

Smoothed Data - Band: Delta, Class: 3
60.5500
61.0924
61.6341
62.1761
62.7197

Smoothed Data - Band: Delta, Class: 4
63.4618
64.0081
64.5721
65.1489
65.7331

Smoothed Data - Band: Delta, Class: 5
45.0779
45.5936
46.1267
46.6738
47.2309

Smoothed Data - Band: Theta, Class: 1
175.1148
176.6032
178.1144
179.6096
181.0551

Smoothed Data - Band: Theta, Class: 2
287.8751
290.3796
292.8610
295.3048
297.7307

Smoothed Data - Band: Theta, Class: 3
201.5067
203.3973
205.2657
207.0897
208.8512

Smoothed Data - Band: Theta, Class: 4
215.9941
216.9570
218.0847
219.4906
221.2501

Smoothed Data - Band: Theta, Class: 5
191.5920
194.1473
196.3510
198.2294
199.8502

Smoothed Data - Band: Alpha, Class: 1
163.4844
164.2648
164.9894
165.7332
166.6172

Smoothed Data - Band: Alpha, Class: 2
144.0401
144.9684
146.1256
147.5732
149.2698

Smoothed Data - Band: Alpha, Class: 3
151.7010
152.2926
152.9502
153.7146
154.6080

Smoothed Data - Band: Alpha, Class: 4
103.3791
104.1915
104.9408
105.6645
106.4184

Smoothed Data - Band: Alpha, Class: 5
282.1970
284.4752
286.1361
287.6289
289.2904

Smoothed Data - Band: Beta, Class: 1
534.1584
536.2628
538.4308
541.9698
546.5168

Smoothed Data - Band: Beta, Class: 2
349.4062
354.3796
359.5033
364.5189
370.1707

Smoothed Data - Band: Beta, Class: 3
283.5573
285.2572
286.5597
287.5619
288.5168

Smoothed Data - Band: Beta, Class: 4
282.8231
285.1619
287.4060
289.4897
291.5041

Smoothed Data - Band: Beta, Class: 5
506.6113
508.8202
510.8172
512.6150
514.4821

part 6

Select a channel (preferably a central channel on the scalp, like CPz), and plot the results from part (e) as follows: For each band, plot the smoothed time-frequency signal (duration: 10 seconds) for each of the 5 classes in a single figure (resulting in a total of 4 figures, each with 5 plots).

solution

the output is shown in figure 16.

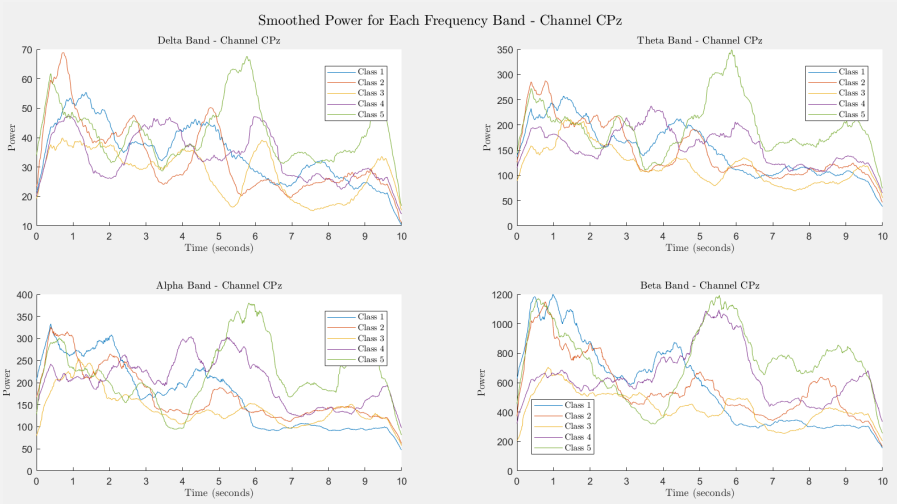

Figure 16. the output of filtering in different bands

part 7

Considering the mental task, starting from the third second, what conclusions can be drawn by comparing the frequency information for each of the five classes?

solution

From the plots provided, we can analyze the smoothed power across different frequency bands (Delta, Theta, Alpha, Beta) for each of the five classes, focusing on the period starting from the third second. Here are some potential conclusions based on the comparison of frequency information:

- **Delta Band**: Generally, the power in the Delta band is lower across all classes as the mental task progresses, with fluctuations that may indicate different levels of engagement or cognitive load. Class 4 shows a more substantial increase around 5–6 seconds, suggesting a distinct response in the Delta band for that specific class.
- **Theta Band**: The Theta band shows a significant rise in power for Class 5 starting around the fifth second. This increase could indicate deeper cognitive processing, as Theta is often associated with memory and attention. Classes 2 and 3 also show higher power, but with different temporal patterns compared to Class 5.

- **Alpha Band**: Alpha band power is typically associated with relaxation or idling mental states. The power decreases for most classes from the third second onward, especially in Classes 2 and 4, possibly indicating increased mental activity or concentration as the task demands increase. Class 5, however, maintains a relatively stable Alpha power, which might suggest a different cognitive response.
- **Beta Band**: The Beta band, often linked to active thinking and focus, shows substantial power increases in Class 1 and Class 5 after the third second, with a peak around 4–5 seconds. This may indicate heightened alertness or engagement in these classes, reflecting active task engagement. The other classes have less pronounced increases, suggesting lower levels of mental focus or a different type of cognitive processing.

Overall Conclusion: Starting from the third second, each class exhibits unique patterns in frequency power across the different bands. For instance:

- **Class 5** shows a strong response in Theta and Beta bands, which could suggest more intensive cognitive processing.
- **Class 4** has a notable rise in the Delta band, possibly indicating a distinct cognitive strategy or state.
- **Class 1** exhibits strong Beta activity, which may reflect focused attention or active cognitive involvement.

These observations suggest that the different classes may represent distinct mental states or task responses, with each frequency band reflecting varying aspects of cognitive load, engagement, and processing style.